b) This question is about threads and concurrency.

[14 marks]

i) Explain how threads are synchronizing in the Java concurrent programming using the ***Mutual exclusion*** and ***Cooperation*** mechanisms.

(2 marks)

ii) Figure 6 below is a Java implementation of Threads in a Java concurrency programming. Explain what the code is doing in lines 3, 11, 24 and 32.

(8 marks)

iii) Predict and write the outputs of the Java program in Figure 6.

(4 marks)

```
1.public class ThreadA {
2. public static void main(String[] args){
3.   ThreadB b = new ThreadB();
4.   System.out.println("Main thread initialising...");
5.     synchronized(b){
6.       b.start();
7.       System.out.println("Thread B started.");
8.       try{
9.          System.out.println("Waiting for Thread B to complete...");
10.         while(!b.finished) {
11.         b.wait();}
12.       }catch(InterruptedException e){
13.        e.printStackTrace();}
14.      System.out.println("Total is: " + b.total);
15.    }
16. }
17.}
18.
19. Public class ThreadB extends Thread{
20.    int total=0;
21.    boolean finished;
22.    @Override
23.    public void run(){
24.      synchronized(this){
25.        try {
26.          for(int i=0; i<5 ; i++){
27.            total += i;
28.          }
29.        }
30.        finally {
31.          finished = true;
32.          this.notify();
33.        }
34.      }
35.    System.out.println("Thread completed.");
36.   }
37. }
```

Figure 5

	Do not v this co
Q2b)	
i) (Basic level)	
Mutual exclusion [NOTE: Any ONE of the following is correct - 1 mark in total] - supported by object locks. [1 mark] - enables multiple threads to independently work on shared data without interfering with each other. [1 mark] **Cooperation** [NOTE: Any ONE of the following is correct - 1 mark in total] - supported by the wait and notify methods of class Object. [1 mark] - enables threads to work together towards a common goal. [1 mark]	
ii) (Medium level)	
Line 3: `ThreadB b=new ThreadB()` : A new Thread of the class `ThreadB` which extends the Java Thread class is created without argument [1 mark]. The instance "b" is created, but it is not yet alive [1 mark]	
Line 11: `b.wait()`: This makes the thread "`b`" to stop execution while it waits [1 mark] for the "`notify`" command to be called or when the time elapses before it can continue execution [1 mark]	
Line 24: `synchronized(this)` : This locks the thread of `ThreadB` [1 mark] to acquire the monitor so that no other thread can compete with it during execution [1 mark]	
Line 32: `this.notify()` : This notifies the thread "`b`" [1 mark] which is waiting in Line 11 to resume execution when the resource is available again. [1 mark]	

iii) (Medium level)
`// Check the order of the output below`
`Main thread initialising...`
`Thread B started.` [1 mark]
`Waiting for Thread B to complete...` [1 mark]
`Thread completed.` [1 mark]
`Total is: 10` [1 mark]	

Solutions

Question 2

a) This question is about threads. **[13 marks]**

i) Describe the FIVE states of a Java Thread. **(6 marks)**

ii) **Figure 3** shows the Java class **`MyCalculation`** where the method **`getNextValue()`** is accessible by two threads. Therefore, the code is not thread-safe. Rewrite and improve the Java code by using an intrinsic lock in **Figure 3** to ensure thread-safe and atomic operation (Include the **`main()`** method to display the output result). **(7 marks)**

```
public class MyCalculation {
  private int value = 0;
  public int getNextValue(){
        return value++;
  }
}
```

Figure 3

Question 2

a) Describe the `Joinpoint` and `Pointcut` terminologies for Spring aspect-oriented programming (Spring AOP). **[4 marks]**

b) This question is about threads. **[21 marks]**

i) Describe the **Runnable**, **Blocked** and **Dead** states of the life cycle of a Java Thread. **(6 marks)**

ii) Write a Java thread program that contains the main Java class called `MainClass` which has the public static main method. Another class in the same program called `ThreadClass` that inherits from a Java Thread class has a `run()` method that is accessible by two threads. The `run()` method prints an integer number from 0 to 49. Use the `synchronized(this)` method within the `run()` method to ensure the threads are safe. `ThreadClass` instance is created within the `MainClass` class under the usual `static main()` method of a Java program. Use `sleep()` to cause the current thread to suspend the execution time for 1000 ms. Your program should contain the `start()`, `synchronized()`, `sleep()` and `notifyAll()` methods at the appropriate locations.

(15 marks)

Question 2

a) This question is about Threads. **[15 marks]**

i) Describe the **Runnable** and **Blocked** states of the life cycle of a Java Thread. **(5 marks)**

ii) Figure 3 below is a Java implementation of Java Thread in a Java program. Explain what the code is doing in lines **3, 4, 9, 20 and 24** respectively. **(10 marks)**

```
1. class ThreadA {
2. public static void main(String [] args) {
3.   ThreadB b = new ThreadB();
4.   b.start();
5.
6.   synchronized(b) {
7.     try {
8.       System.out.println("Waiting for b to complete...");
9.       b.sleep();
10.    } catch (InterruptedException e) {}
11.    System.out.println("Total is: " + b.total);
12.  }
13. }
14. }
15.
16. class ThreadB extends Thread {
17.   int total;
18.
19.   public void run() {
20.     synchronized(this) {
21.       for(int i=0;i<100;i++) {
22.         total += i;
23.       }
24.       notifyAll();
25.     }
26.   }
27. }
```

Question 2

a) This question is about Threads. [14 marks]

i) Describe the Runnable and Running states of the life cycle of a Java Thread. (4 marks)

ii) Figure 3 below is a Java implementation of Java Thread in a Java program. Explain what the code is doing in lines **3, 4, 9, 20 and 24** respectively. (10 marks)

```
1. class ThreadA {
2. public static void main(String [] args) {
3.   ThreadB b = new ThreadB();
4.   b.start();
5.
6.  synchronized(b) {
7.    try {
8.      System.out.println("Waiting for b to complete...");
9.      b.wait();
10.   } catch (InterruptedException e) {}
11.   System.out.println("Total is: " + b.total);
12.  }
13. }
14. }
15.
16. class ThreadB extends Thread {
17.   int total;
18.
19.   public void run() {
20.     synchronized(this) {
21.       for(int i=0;i<100;i++) {
22.         total += i;
23.       }
24.       notify();
25.     }
26.   }
27. }
```

Figure 3: A Java Code with Thread

Solutions

	Do not write in this column
Q2ai)	
Runnable: This is when the Thread instance variable invokes the "start()" method eg using the "t.start()" [1 mark] command, where t is the thread instance variable. The thread is now alive and has joined the queue of threads about to be executed [1 mark]	
Running: This is when the Thread has got its chance leaving the queue to access the resource [1 mark] to execute and is now executing [1 mark]	
Q2aii)	
Line 3: ThreadB b=new ThreadB(): A new Thread of the class ThreadB which extends the Java Thread class is created without argument [1 mark]. The instance "b" is created, but it is not yet alive [1 mark]	
Line 4: b.start(): The new Thread created is started using the start() method to become alive [1 mark] and qualify to join the queue [1 mark]	
Line 9: *b.wait():* This makes the thread "b" to stop execution while it waits [1 mark] for the "notify" command to be called or when the time elapses before it can continue execution [1 mark]	
Line 20: *synchronized(this):* This locks the thread of ThreadB [1 mark] to acquire the monitor so that no other thread can compete with it during execution [1 mark]	
Line 24: *notify():* This notifies [1 mark] the thread "b" which is waiting in Line 4 to resume execution when the resource is available again. [1 mark]	
	14 marks

Tutors

c) Describe an **Interrupt and give ONE example how the interrupt methods are invoked** in a Java program. Also describe the THREE ways to terminate a Java ***Thread***.

[7 marks]

	Do not write in this column
Q2c	
An interrupt is an indication to a thread that it should stop what it is doing [1 mark] and do something else [1 mark]. For example it is always implemented in Java using the Thread.interrupted()[1 mark] or t.interrupt() (where t is an instance of a thread class and interrupted() or interrupt() are methods in Java) commands [1 mark]	
THREE ways to terminate a Thread are:	
• The thread has finished the work it is designed to do, and exits the run() method naturally. .[1 mark]	
• The thread has received an interruption signal, while doing its work. 1 mark]	
• The thread has been marked as daemon thread and when the parent thread who created this thread terminates, this thread will be forcefully terminated by the JVM [1 mark]	
	7 marks

Solutions

Question 2

a) This question is about Threads.

[15 marks]

i) With the help of a clearly labelled diagram, explain the FIVE stages and states in the life cycle of a Thread.

(10 marks)

ii) The Java code in Figure 2 describes how a typical Java Thread works in a programme. Explain what the code is doing in lines **5, 6, 7, 10** and **12** respectively.

(5 marks)

```
1. import java.util.*;
2. public class MySynchronizedYield implements Runnable {
3.   public static void main(String[] args) {
4.         MySynchronizedYield r=new MySynchronizedYield();
5.         Thread t=new Thread(r);
6.         t.start()
7.         synchronized(t) {
8.             try {
9.               System.out.println("Please wait for me....");
10.              t.wait(100);
11.                  System.out.println("Please yield for another....");
12.              t.yield();
13.             } catch (InterruptedException e) {}
14.        }
15.   }
16. }
```

Figure 2: A Java Code with Thread

c) ***Visibility*** and ***Race Conditions*** are two problems that may occur when storage of data happens in different memory areas of the computer memory model as Java threads attempt to make changes to the data Using your knowledge of the Java memory model and computer memory model, describe TWO solutions you may use in your multi-threaded Java programme to prevent the problems of ***Visibility*** and ***Race Conditions***.

[4 marks]

	Do not write in this column
	4 marks

d) Define ***Interrupt*** and describe THREE ways to terminate a ***Thread***.

[4 marks]

	Do not write in this column